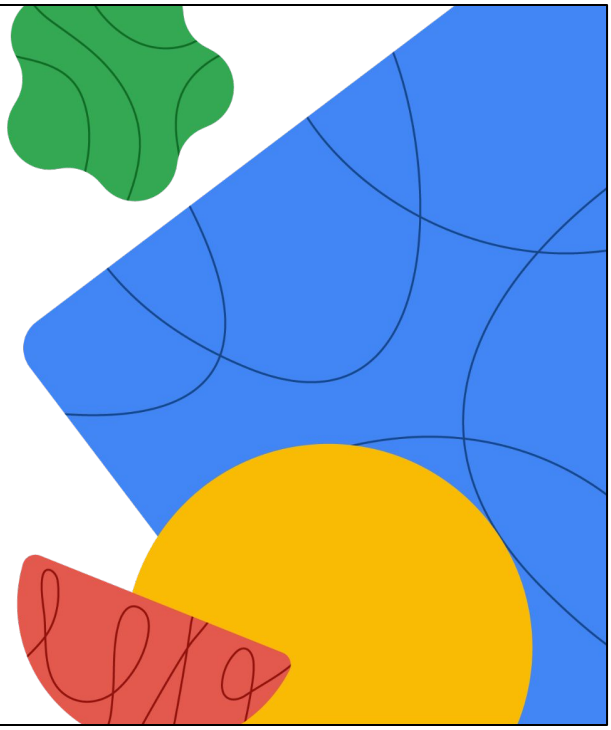
Understanding LookML and the semantic layer

Introduction to Looker



Let’s explore LookML, Looker's data modeling language, and its role in creating a semantic layer for consistent and reliable data analysis.

In this module, you ...

In this module, you:

In this module, you ...

01 Understand why Looker developed LookML.

- Understand why Looker developed LookML.

In this module, you ...

01	Understand why Looker developed LookML.
02	Define LookML and the semantic layer.

- Define LookML and its role in creating a semantic layer.

In this module, you ...

01	Understand why Looker developed LookML.
02	Define LookML and the semantic layer.
03	Describe the benefits of using LookML for data consistency and governance.

- And describe the benefits of using LookML for data consistency and governance.

01 LookML and the semantic layer

02 Data governance in Looker

Now, let's delve deeper into LookML and the semantic layer, and explore how they can transform your data analysis workflow.

```
view: orders {
   dimension: order_id {
      primary_key: yes
      Type: number
}

dimension: order_date {
   Type: date
}

dimension: customer_id {
   Type: number
}

measure: total_amount {
   type: sum
   sql: ${TABLE}.amount; ;
}
```

LookML

LookML is Looker's own data modeling language

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LookML

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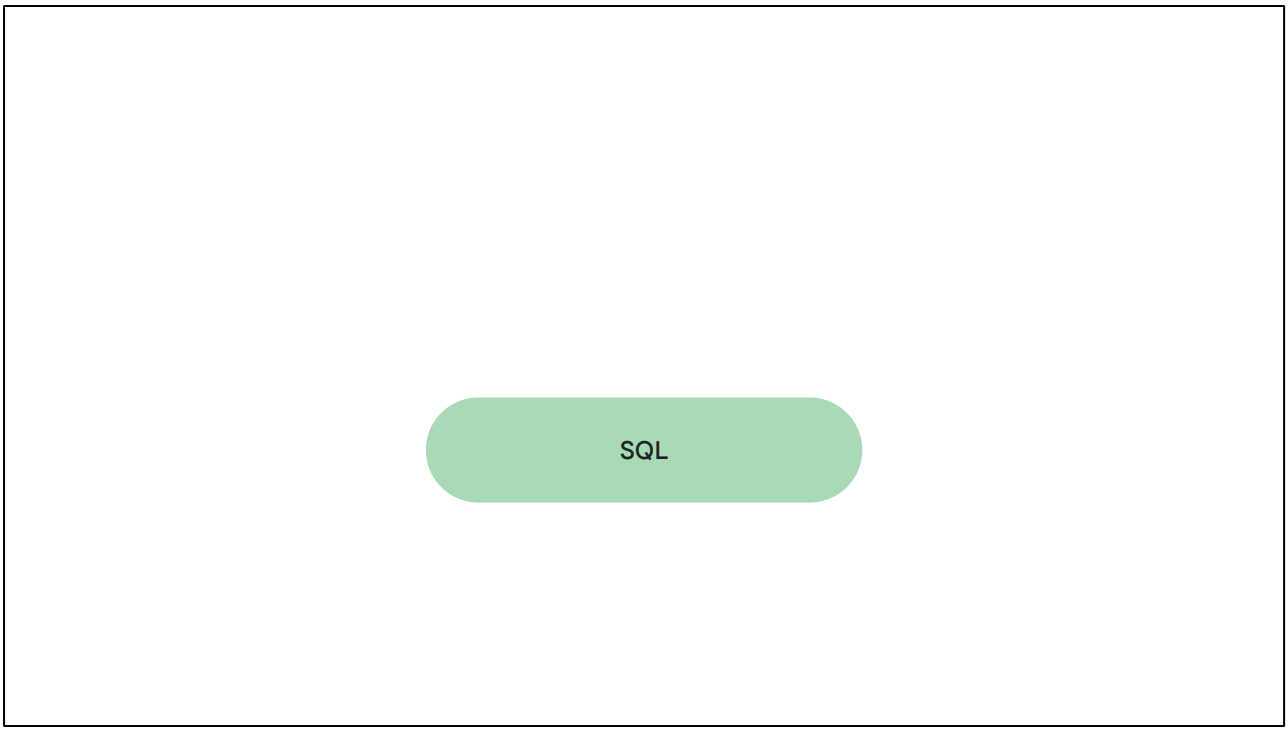
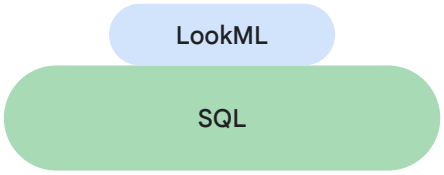
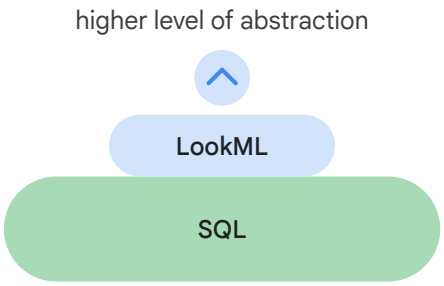
SQL. SQL, or Structured Query Language, is a standard language for communicating with databases. Don't worry. Even if you're not an SQL expert, you can still learn and use LookML effectively.



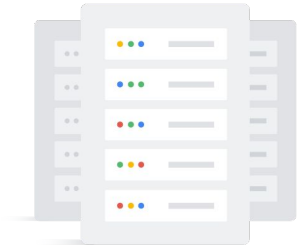
Essentially,



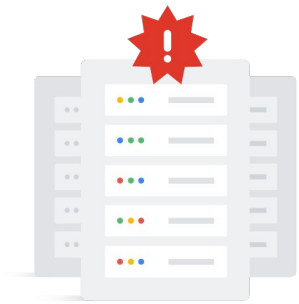
LookML provides



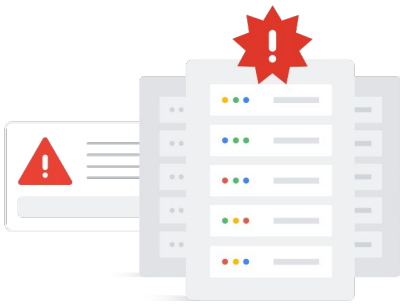
a higher level of abstraction on top of SQL. It allows you to define data elements and relationships in a more business-oriented way, without having to write complex SQL queries directly.



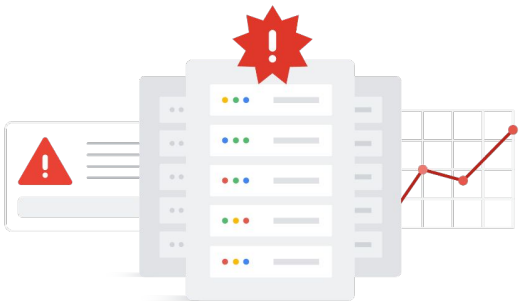
Why did Looker develop LookML in the first place?



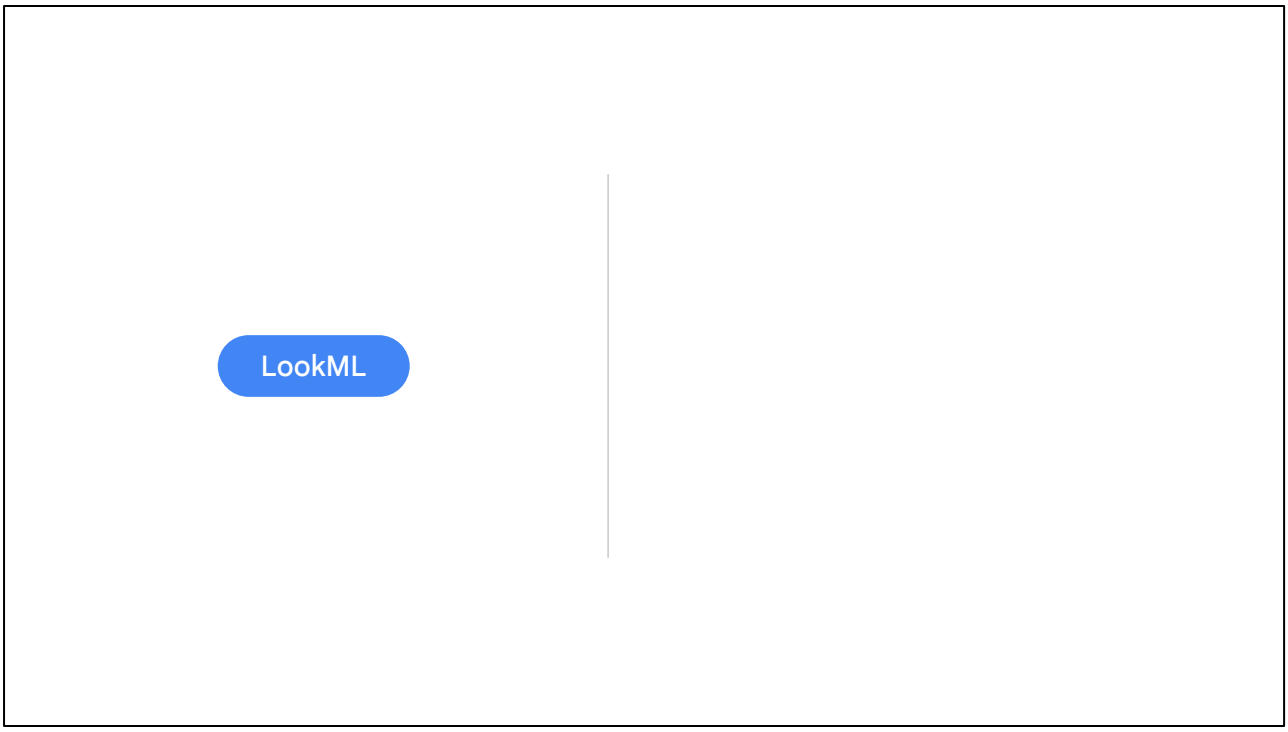
Because of the challenges of working with raw data.



Directly querying a database can be complex and error-prone,



even for experienced data analysts. Different teams might have different interpretations of the same data, leading to conflicting reports and confusion.

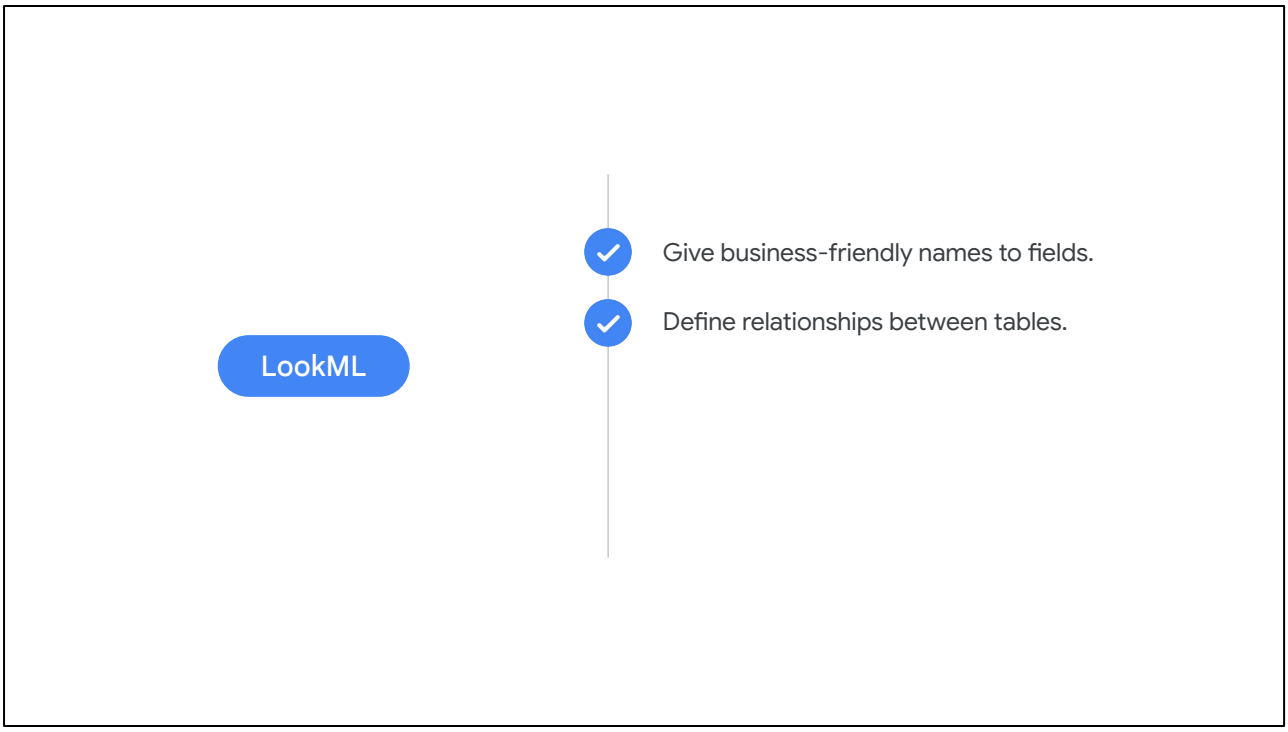


LookML addresses these challenges by creating a single source of truth for your data. It allows you to:

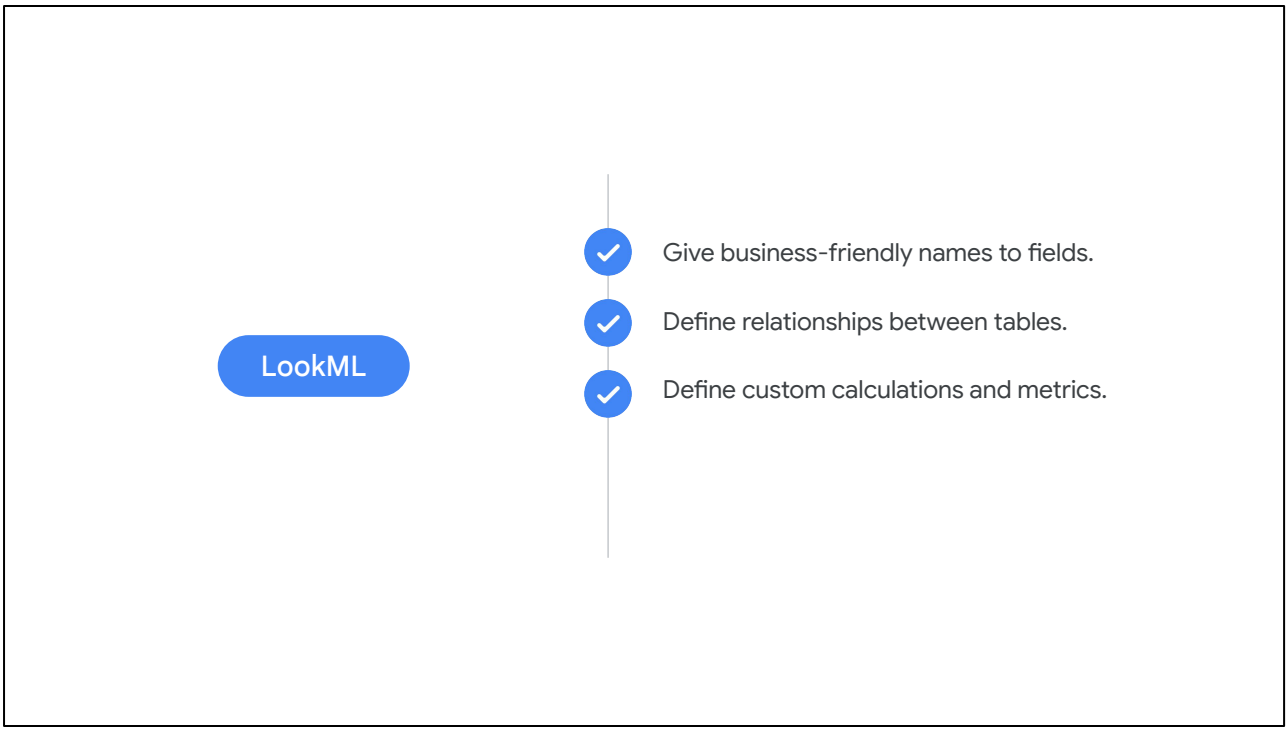
LookML

Give business-friendly names to fields.

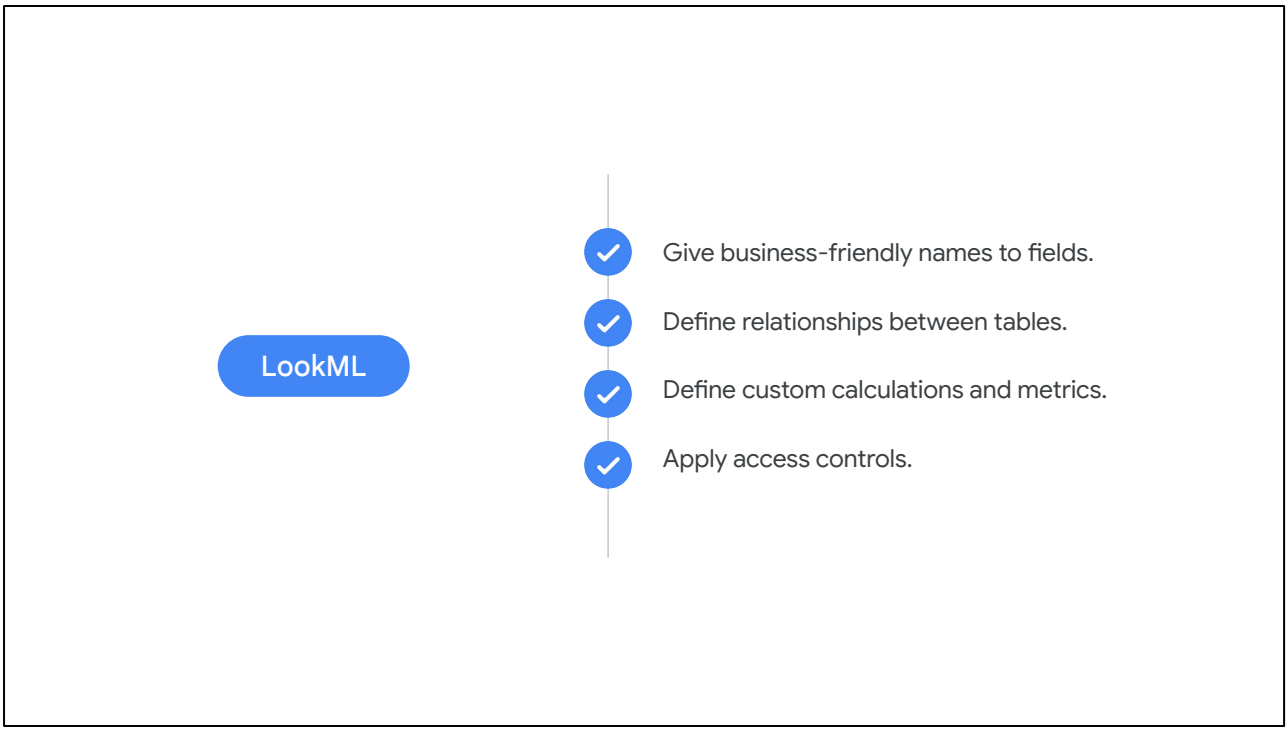
- Give clear, business-friendly names to database fields.



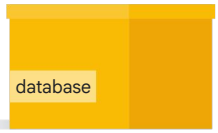
- Define relationships between tables, making it easier to understand how data is connected.



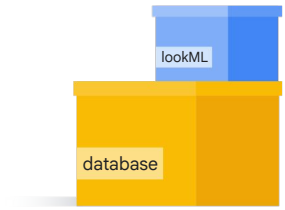
- Define custom calculations and metrics that are relevant to your business.



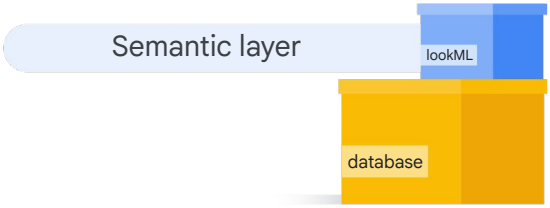
- And apply access controls, ensuring that sensitive data is only visible to authorized users.



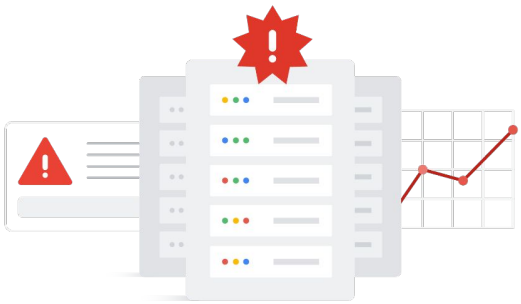
This curated layer of data definitions and logic



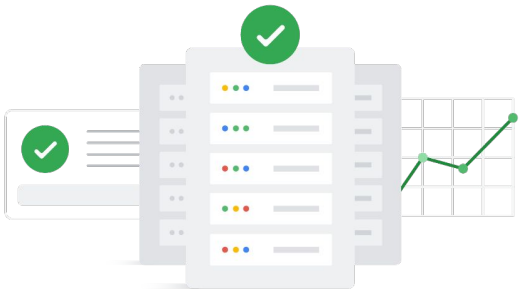
built on top of your database is what we call



the semantic layer. It's like a map that guides everyone in your organization to the same data destination, ensuring consistency and clarity.



With LookML and the semantic layer, you can say goodbye to data silos, conflicting reports, and confusion about metrics.

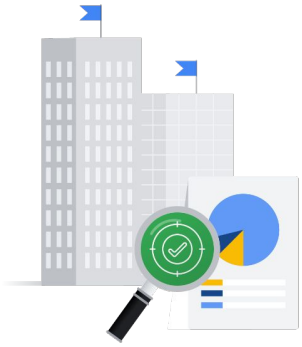


Everyone in your organization can work with the same data, the same definitions, and the same understanding.

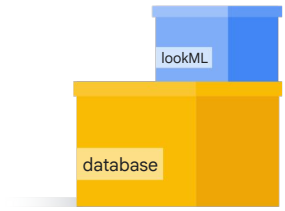
01 LookML and the semantic layer

02 Data governance in Looker

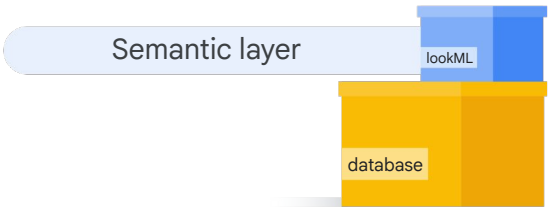
In this lesson, we dive deeper into how Looker empowers you to manage and govern your data effectively.



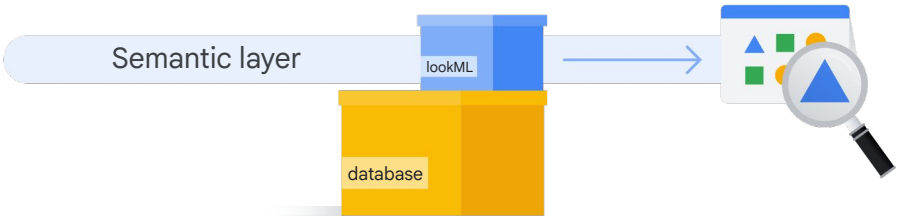
Imagine you're a data analyst at a large retail company. You're responsible for ensuring that data is accurate, consistent, and accessible to the right people across different departments, from marketing to sales to inventory management. This is where Looker's data governance features come into play.



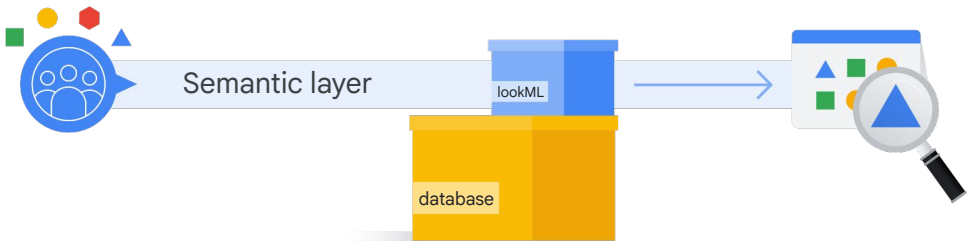
LookML, with its ability to define clear data definitions and relationships, plays a crucial role in establishing and maintaining a robust data governance framework within Looker.



The semantic layer, built with LookML,

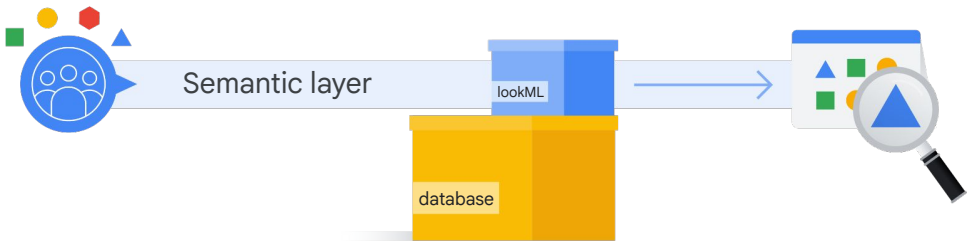


acts as a central source of truth for data definitions,



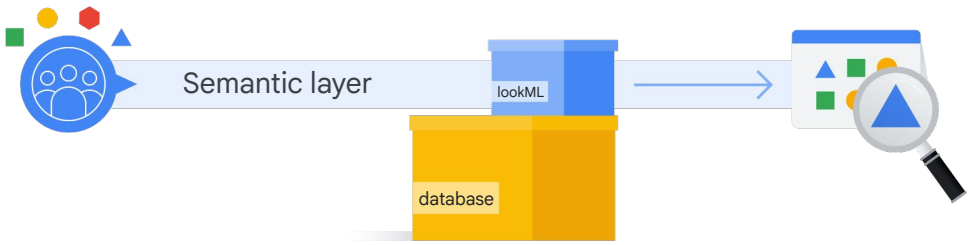
ensuring that everyone in your organization is working with the same data understanding.

This consistency is essential for effective data governance, as it enables



Accurate reporting

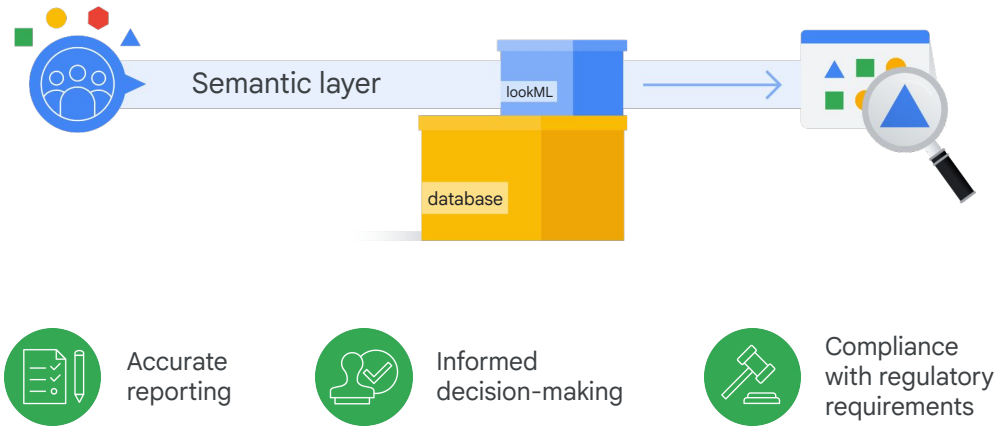
accurate reporting,



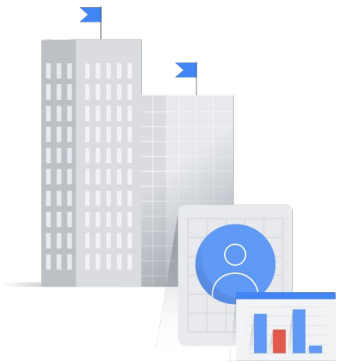
Accurate reporting

Informed decision-making

informed decision-making,

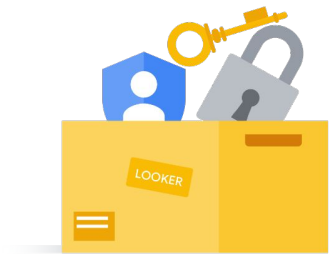


and compliance with regulatory requirements.



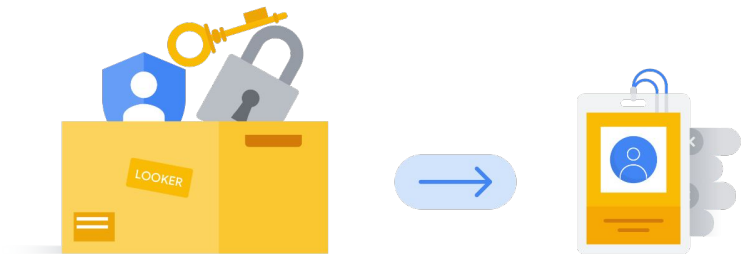
Now, let's return to our retail company data analyst scenario to explore the specific ways Looker empowers you to govern your data and apply it to your own work:

Access controls



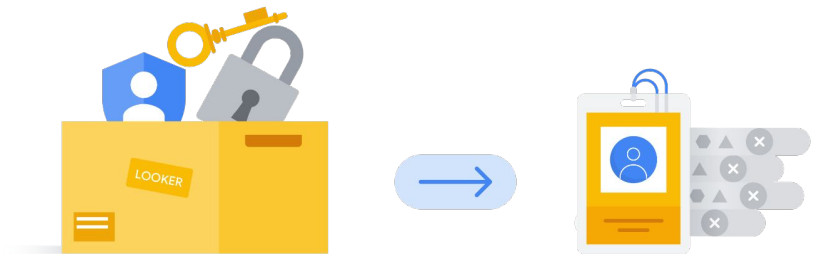
Access controls:

Access controls



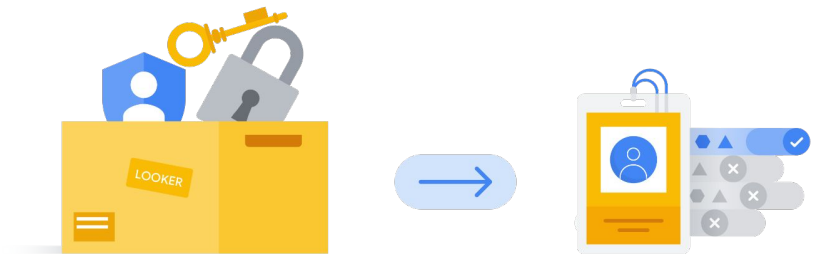
Looker provides a robust set of access controls

Access controls



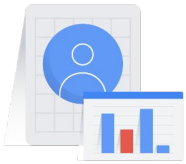
to help you define and enforce data governance policies.

Access controls



You can control access to sensitive data, ensuring that only authorized users can view or modify it.

Access controls

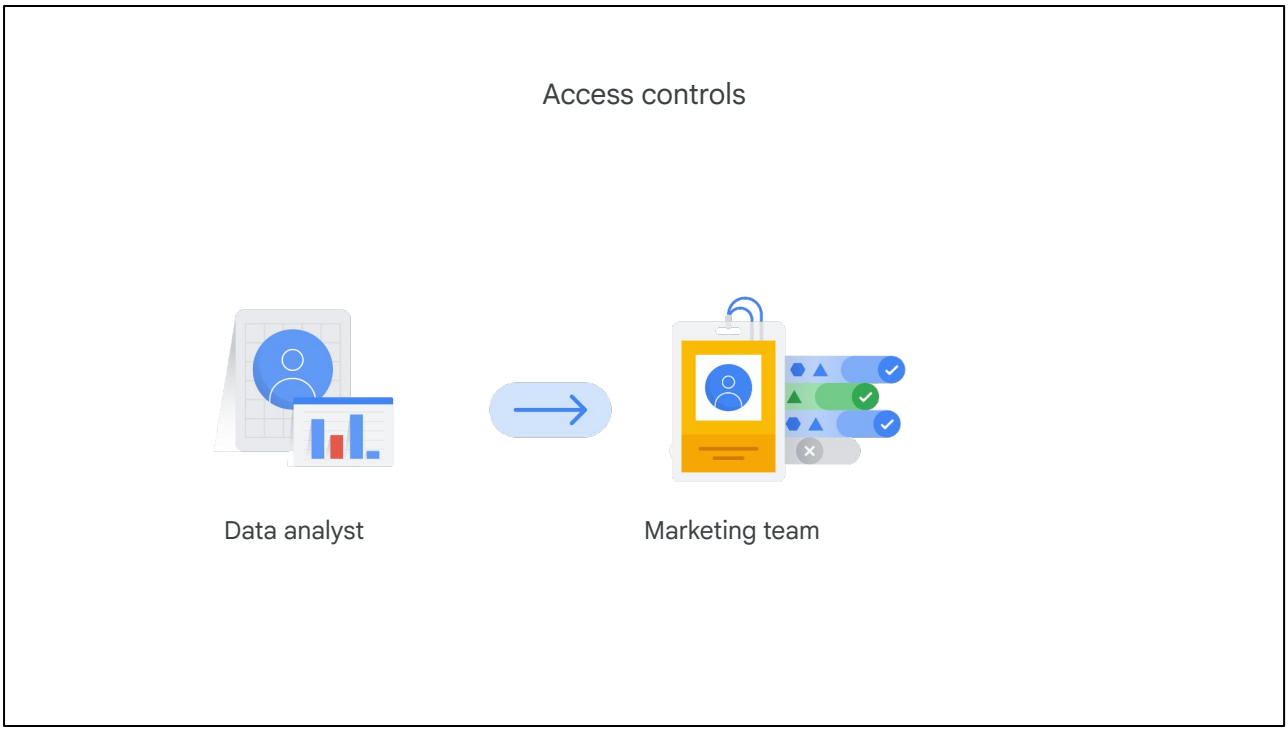


Data analyst

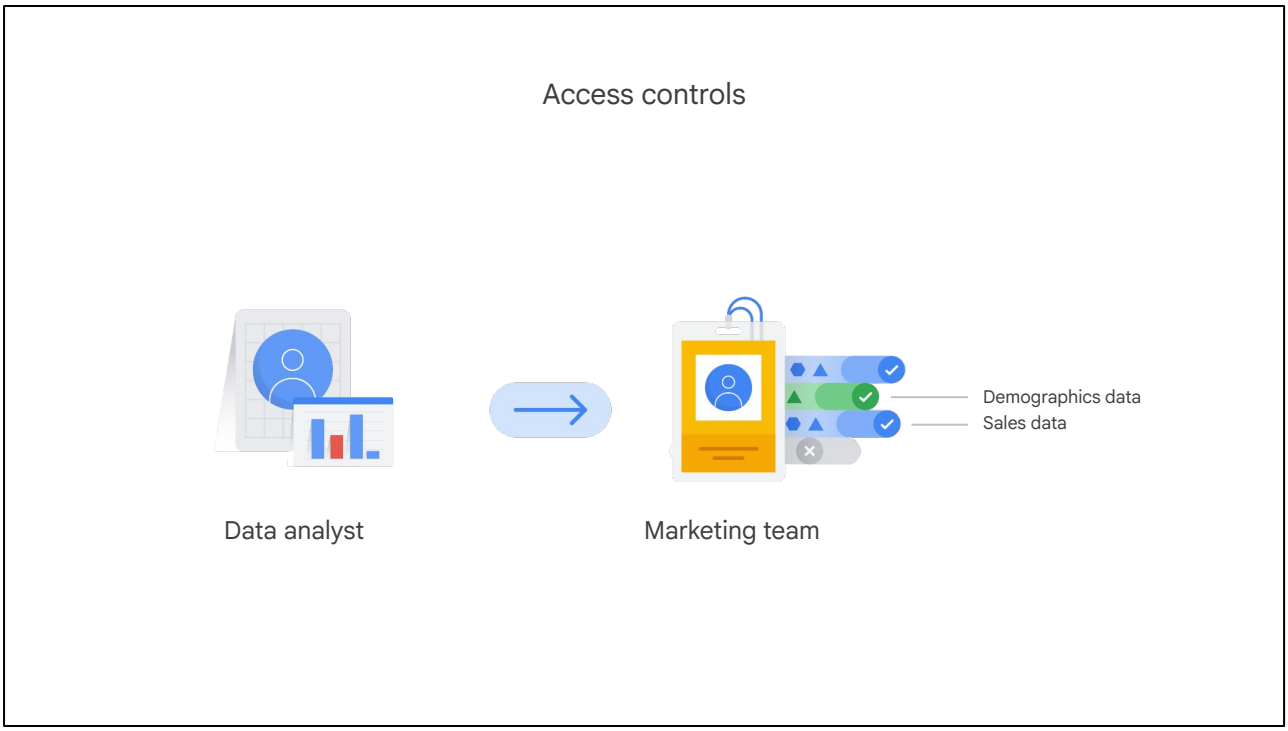
As the retail company's data analyst,

Access controls

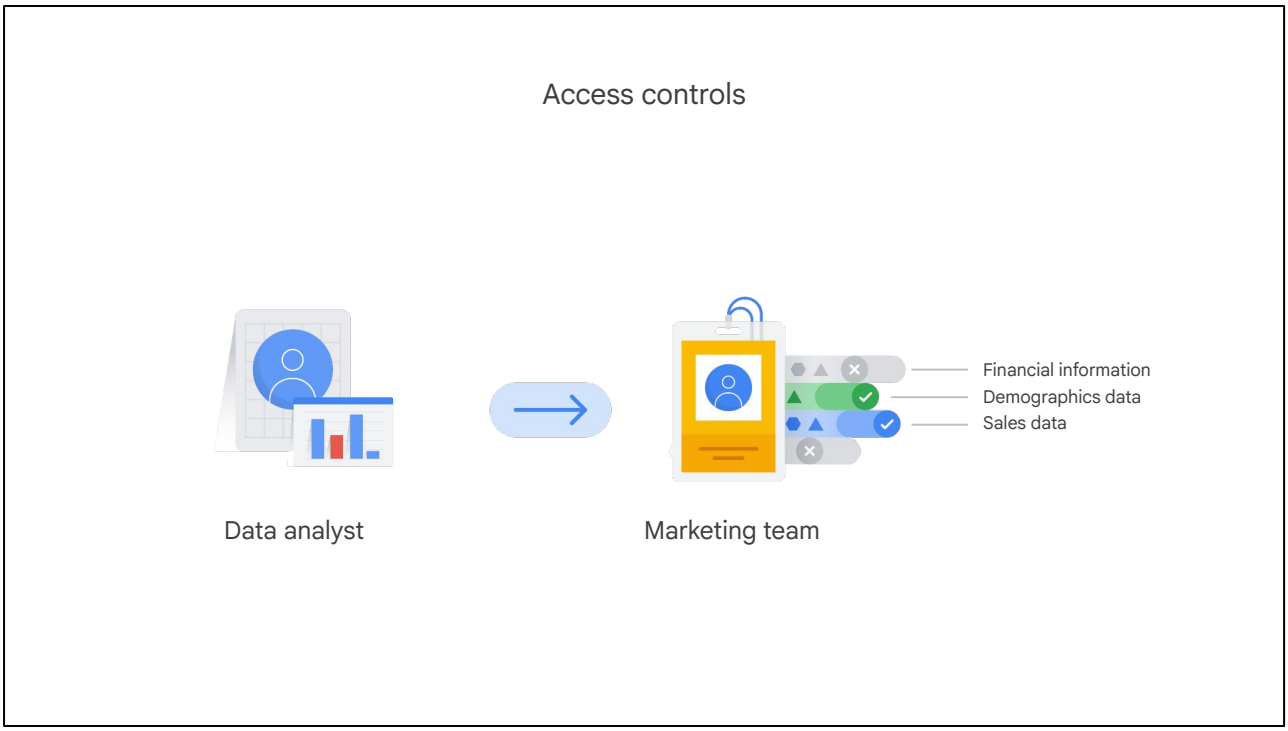
you can use Looker's access controls to define user roles and permissions, specifying who can access which data and what actions they can perform.



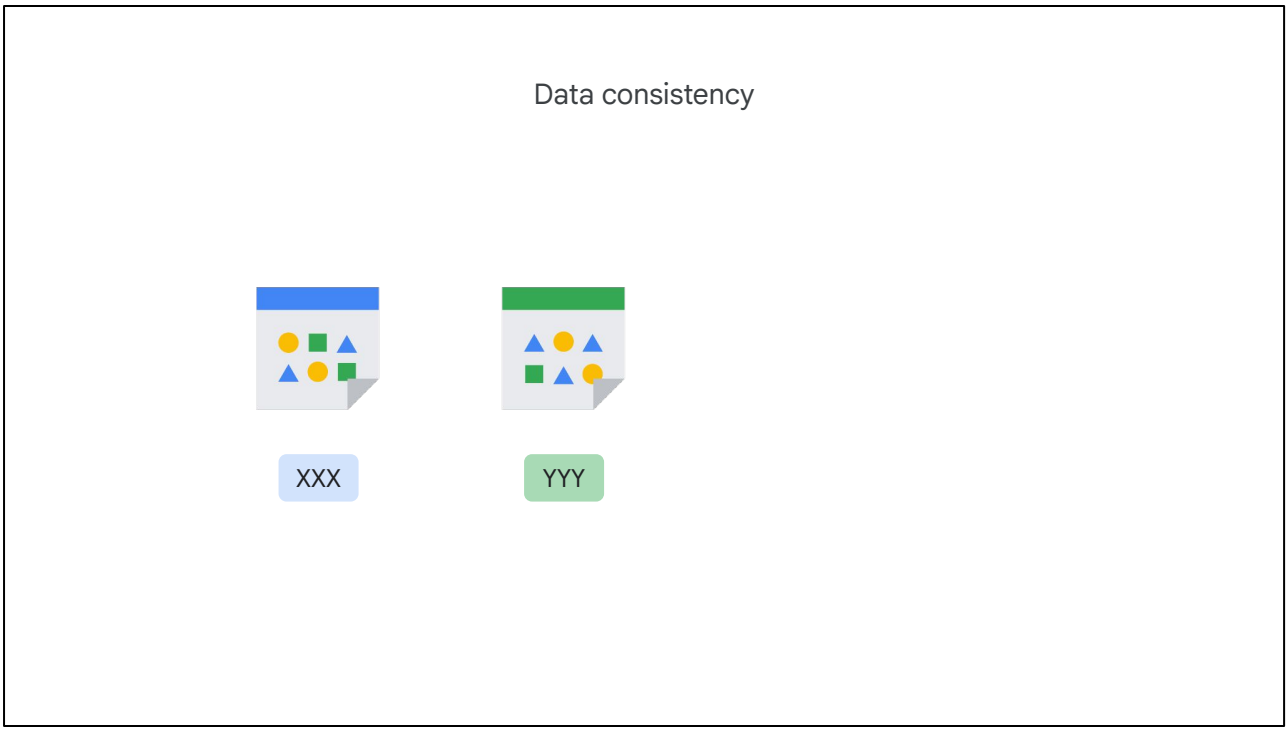
For example, you might grant marketing team members access to



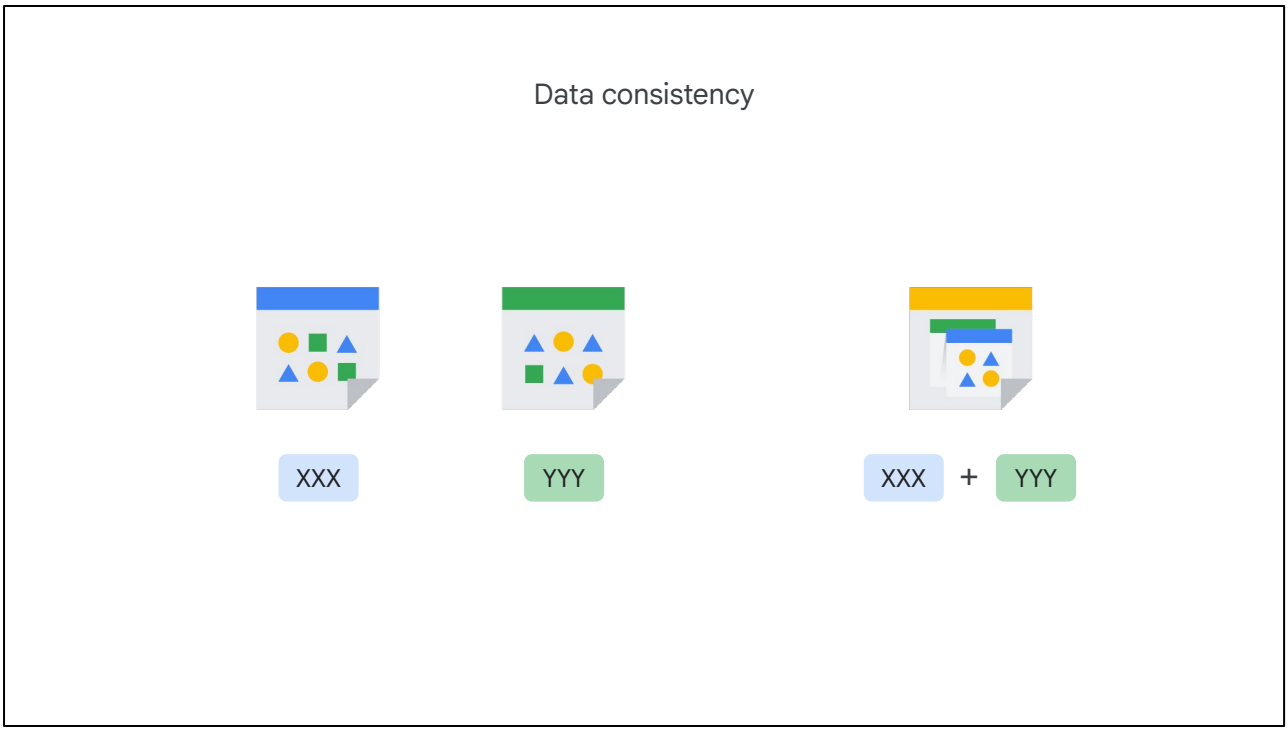
customer demographics and sales data,



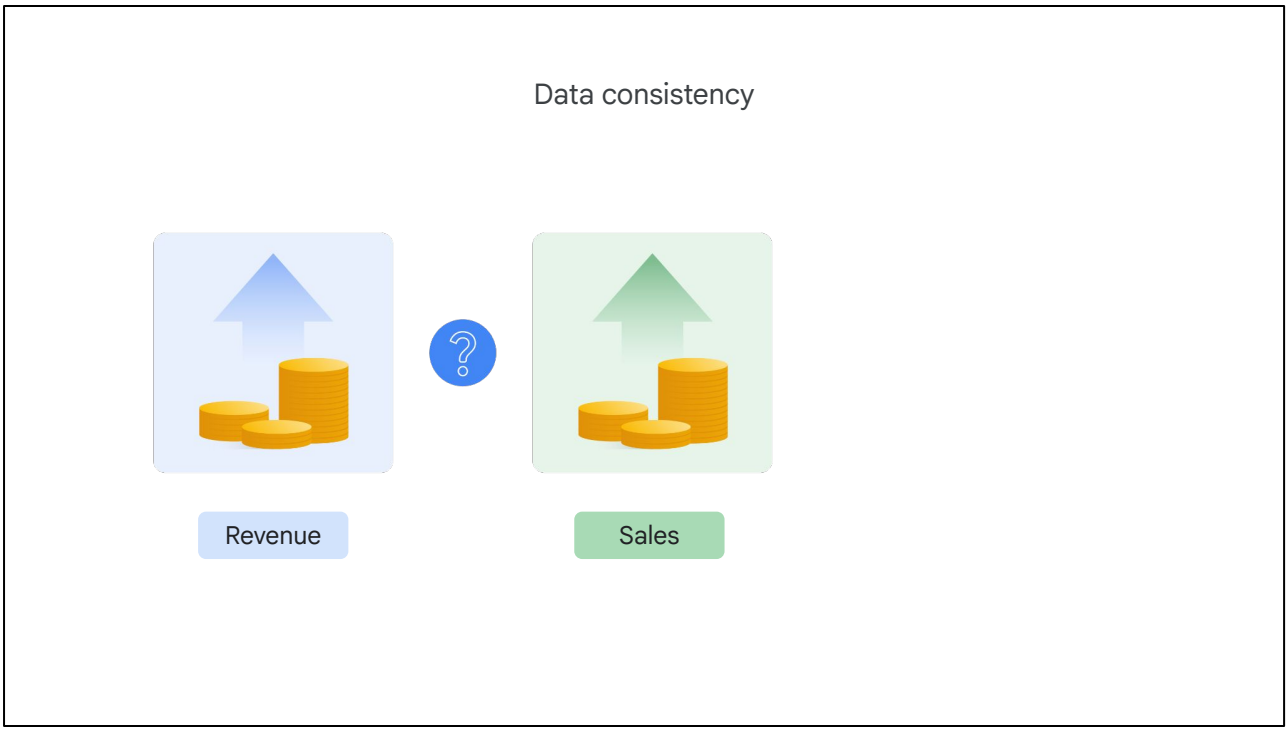
while restricting access to sensitive financial information.



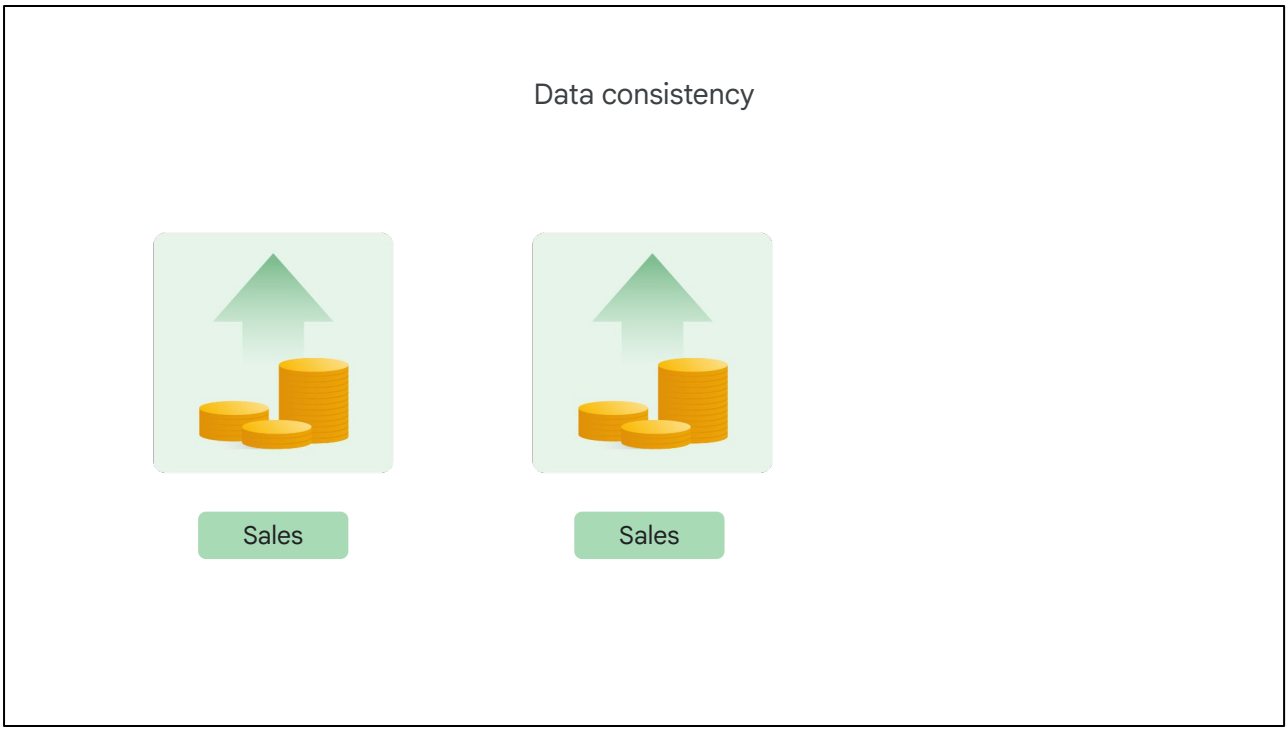
Data consistency: Looker helps you maintain data consistency by enforcing naming conventions.



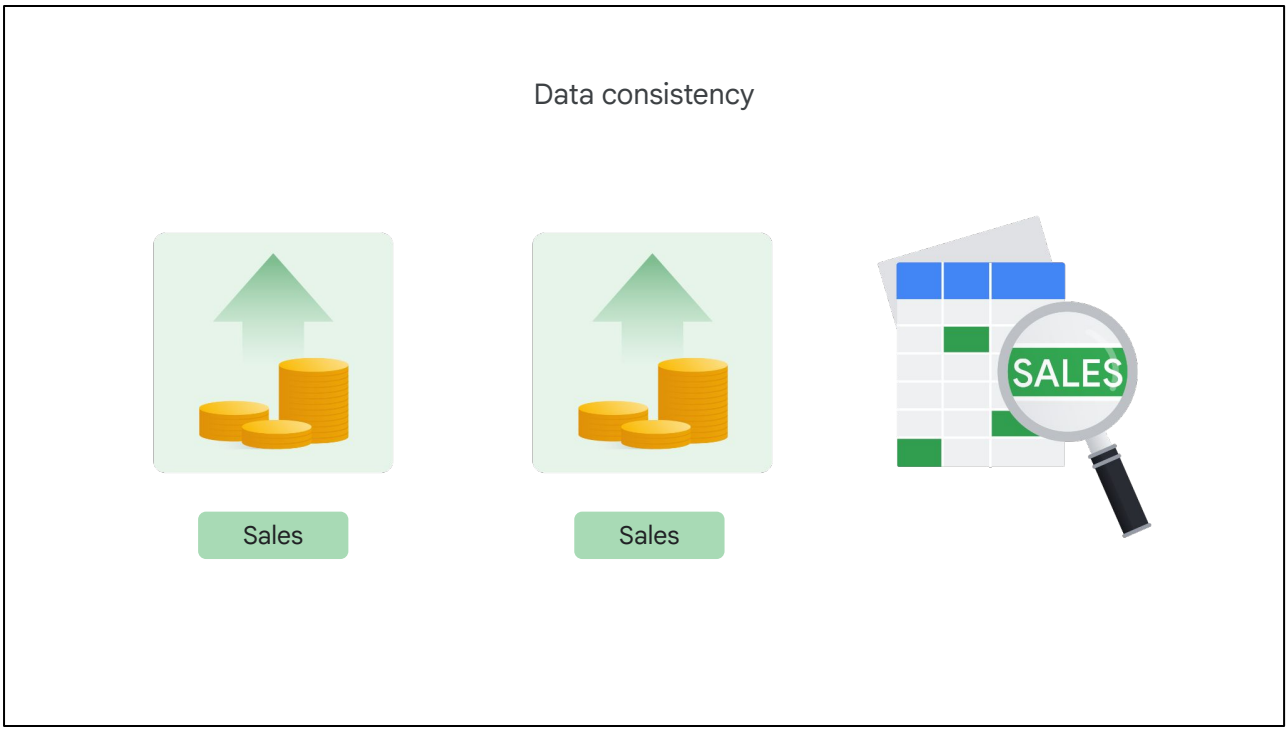
You can define clear rules for how data should be named and organized, ensuring that everyone in your organization is on the same page.



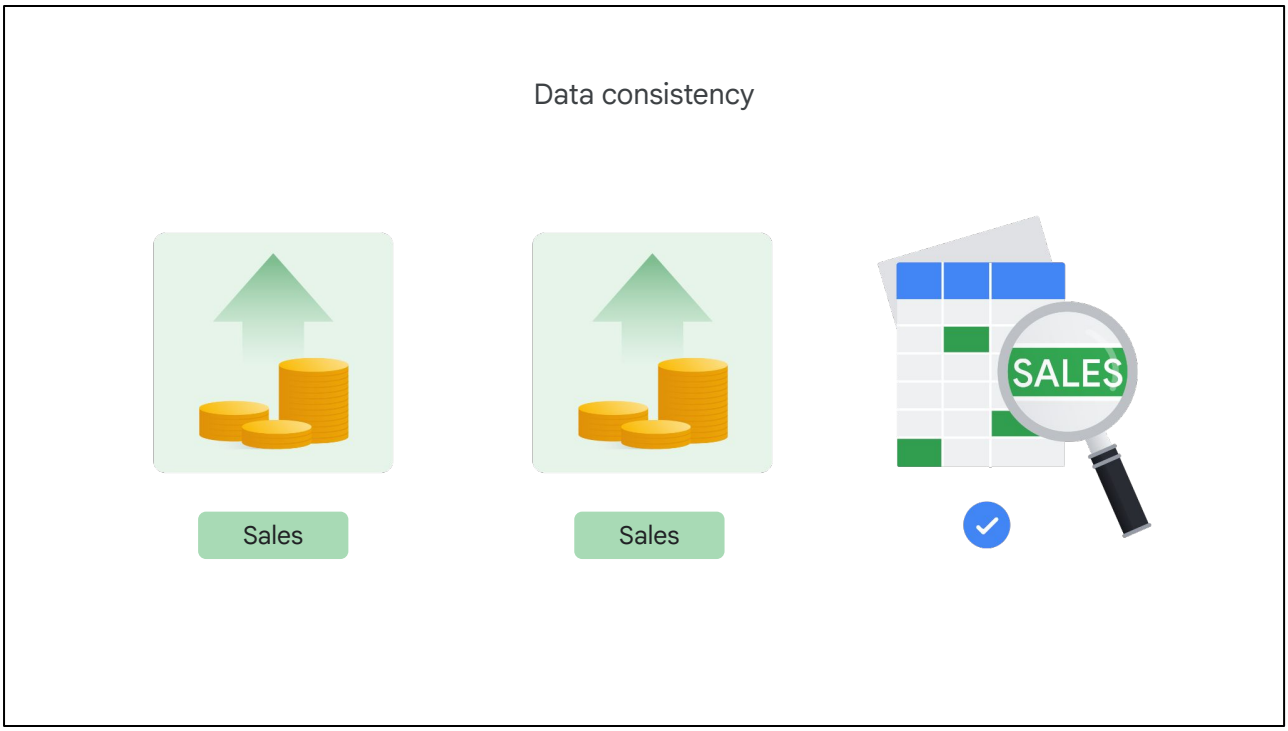
In our retail scenario, imagine defining a clear naming convention for measures, ensuring that everyone uses the same terminology. For example, whether to use "Revenue" or



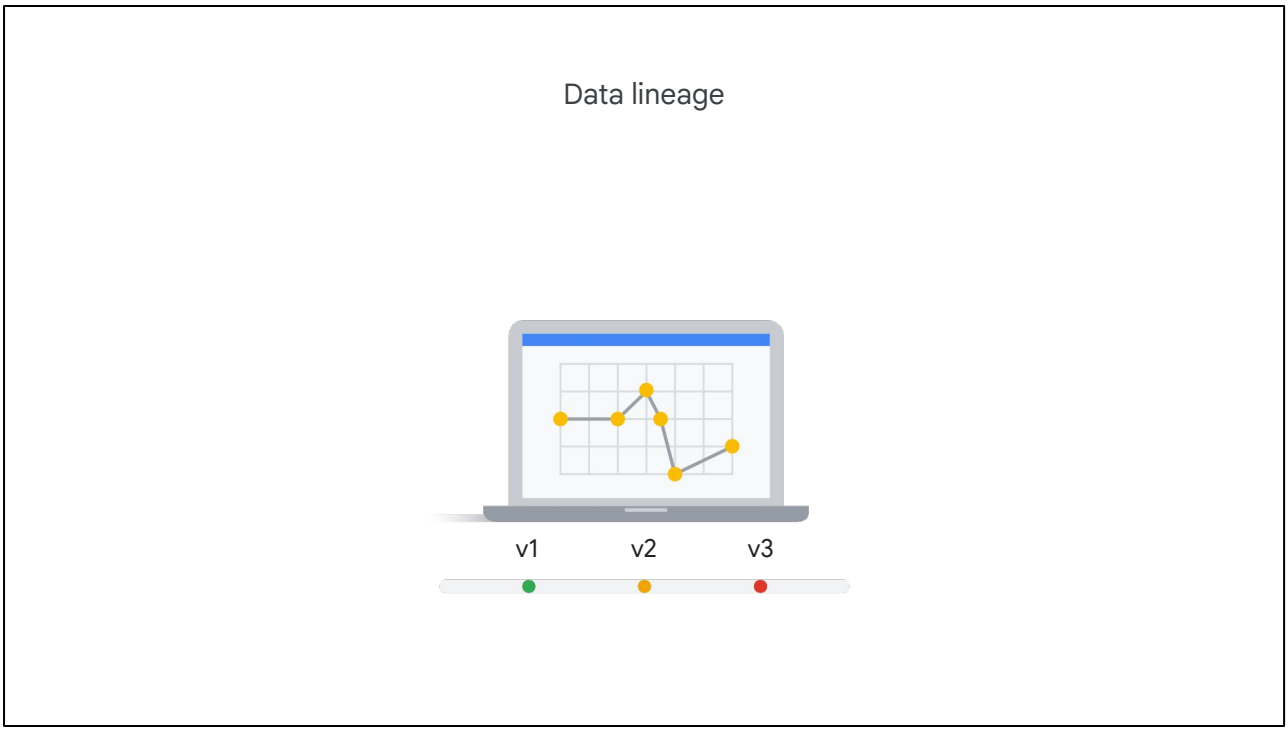
"Sales" consistently throughout the company.



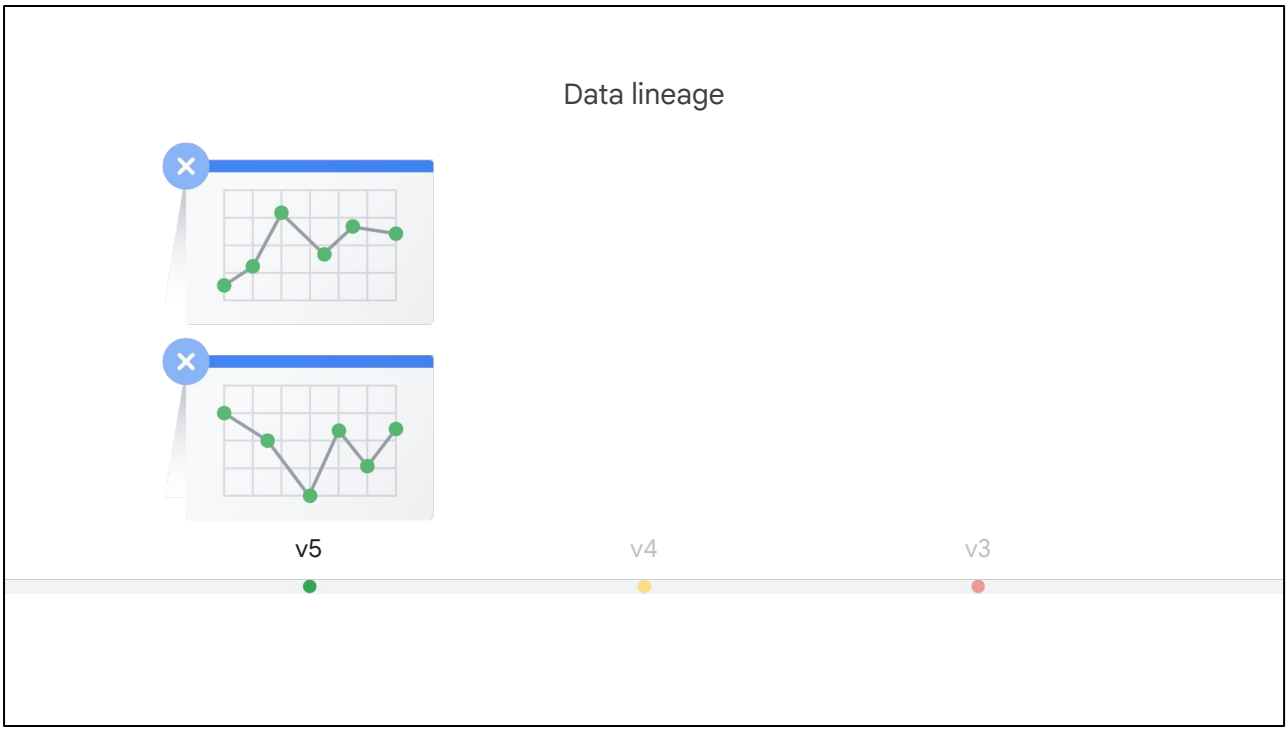
This consistency makes it easier



to analyze data across different departments and reports.



Data lineage: Data lineage tracking allows you to trace the origin and transformations of your data, providing a clear understanding of how data flows through your system. This helps you identify and resolve quality issues, ensuring that your data is accurate and reliable.



For instance, if you notice a discrepancy in sales figures between two reports,

you can use data lineage to track the source of the data

and identify any transformations or calculations

that might have caused the discrepancy.

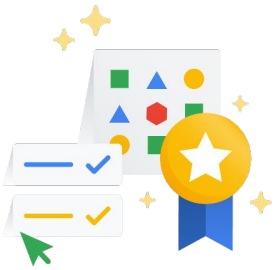
Data validation and testing

Data validation and testing: Looker also provides tools for data validation

Data validation and testing

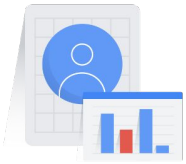
and testing,

Data validation and testing



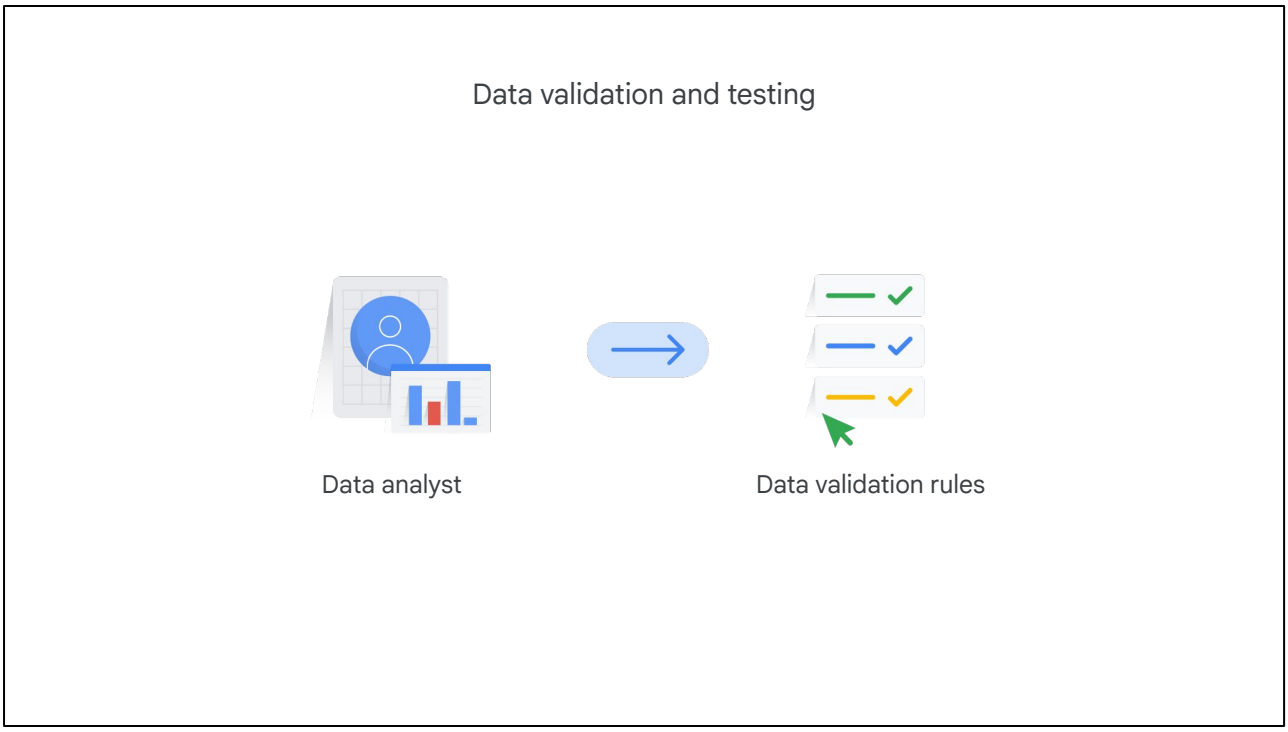
allowing you to define rules and checks to ensure data quality. You can automatically validate data against predefined criteria, identifying and flagging any inconsistencies or errors.

Data validation and testing

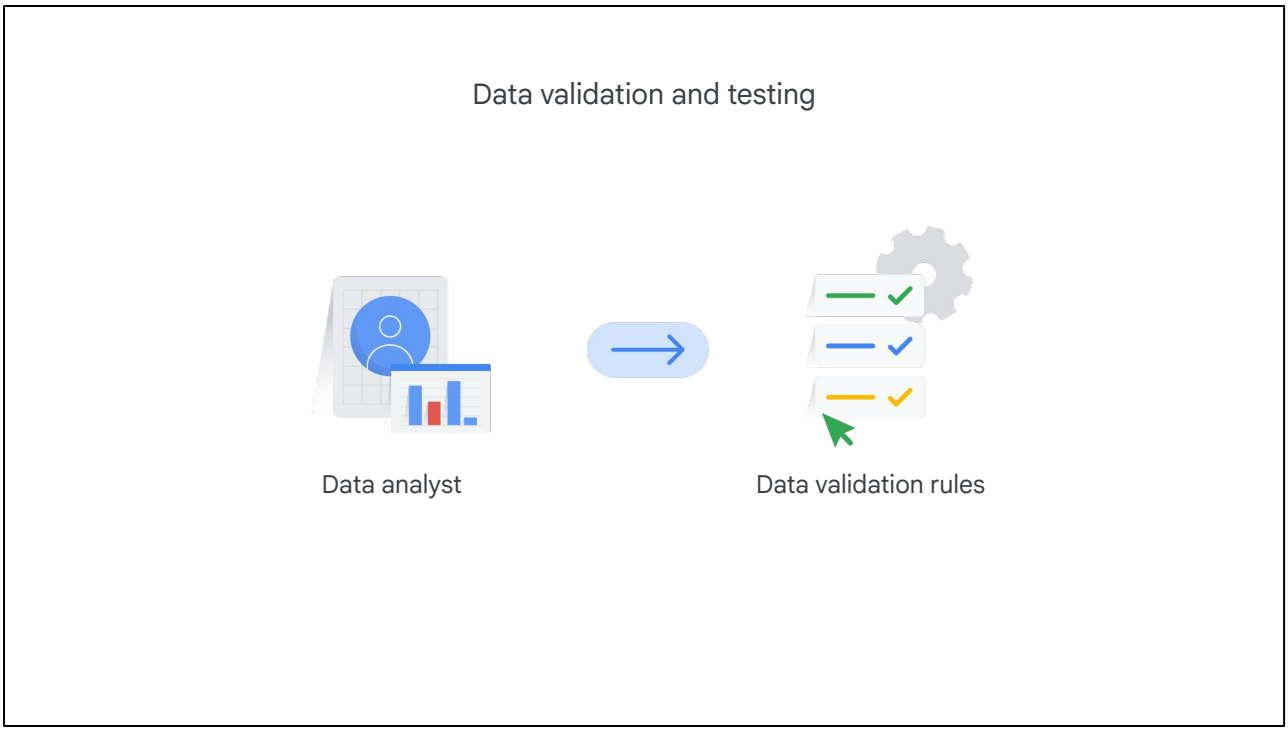


Data analyst

As the data analyst, you could set up



data validation rules to ensure

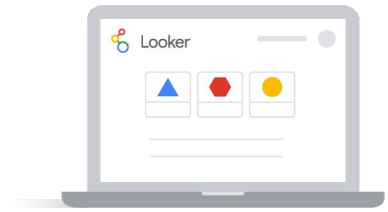


that product prices are always positive values, or that customer ages are within a reasonable range.

Data validation and testing

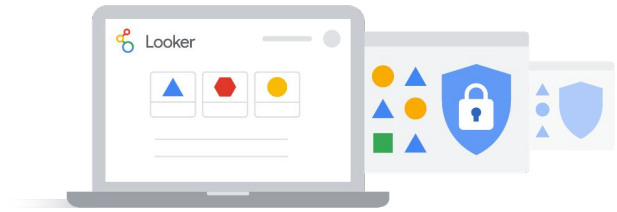
This helps you proactively identify and address potential data quality issues.

Integration with other tools



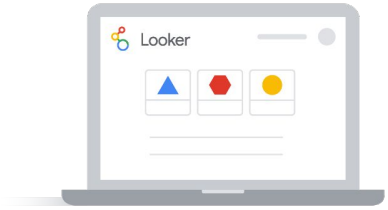
Integration with other tools: Additionally, Looker integrates

Integration with other tools



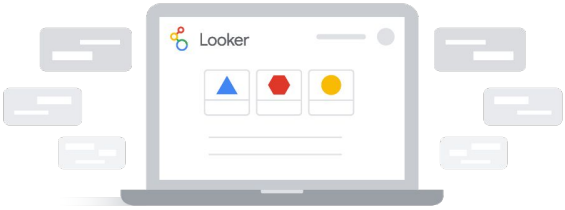
with other data governance tools and platforms, allowing you to

Integration with other tools



seamlessly incorporate Looker

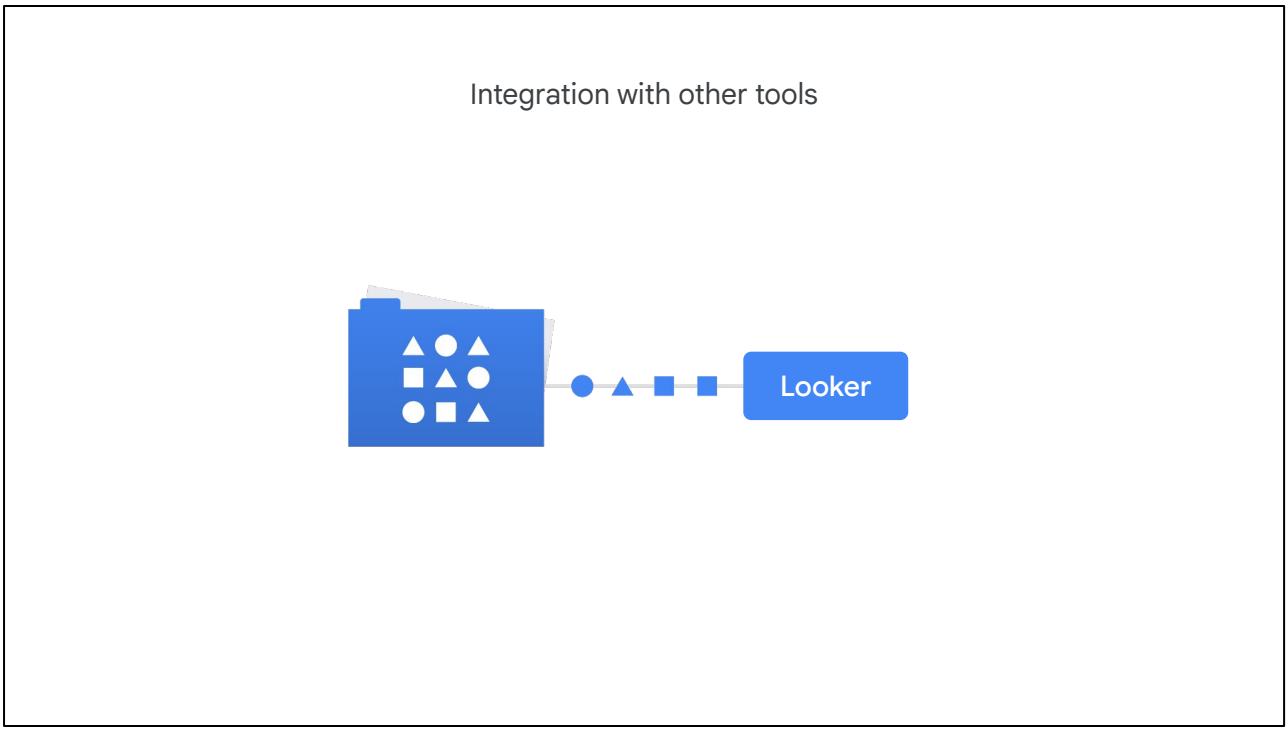
Integration with other tools



into your existing data governance framework.

Integration with other tools

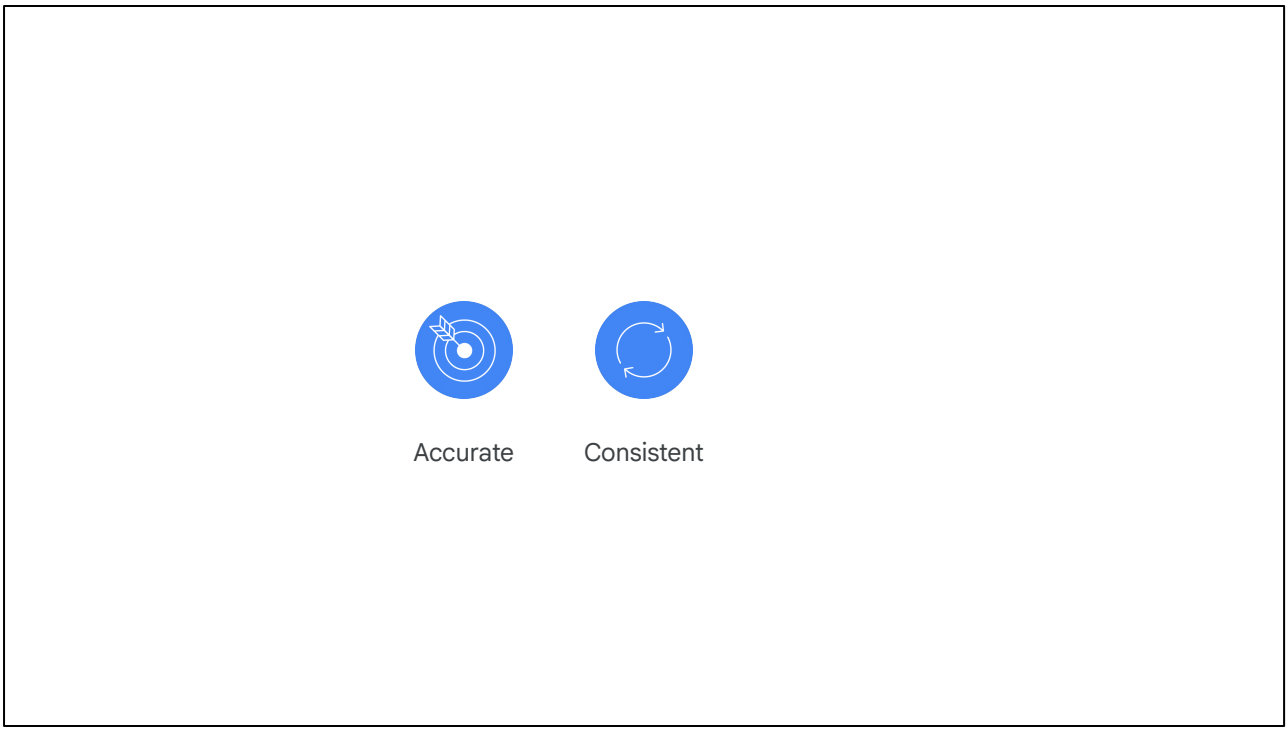
For example, if your company uses a separate data catalog or data quality tool,



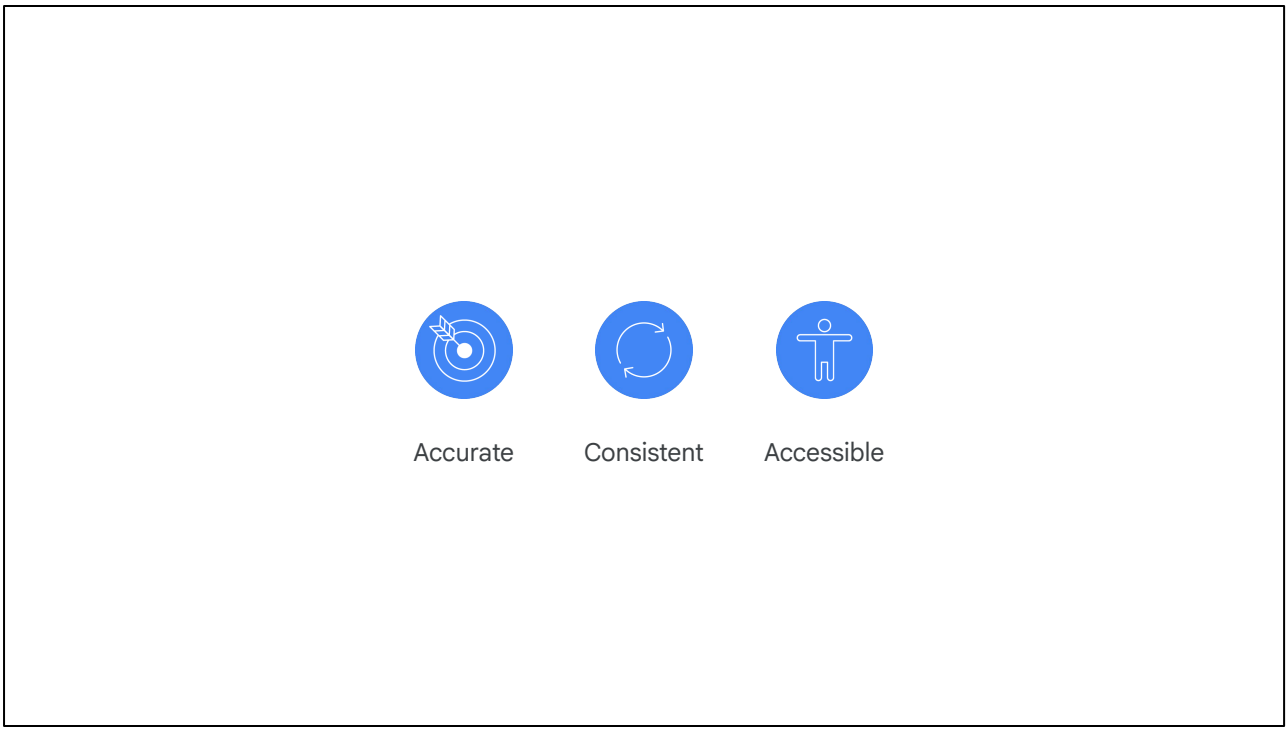
you can integrate Looker with those tools to streamline your data governance processes. Given this, Looker's data governance features help ensure your data is

Accurate

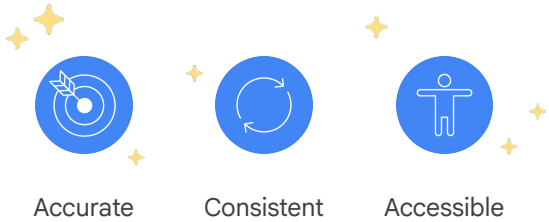
accurate,



consistent,



and accessible to the right people.



This empowers your organization to make informed decisions and drive business success.